gpeyt1-18 · Mononobe–Okabe

GEO PE · Long Calculation Problems I — Topic 4: Earthquake Engineering & Dynamic Loads

Problem

An 18 ft cantilever wall retains level, dry granular backfill (γ = 120 pcf, φ = 34°, c = 0). The back face is vertical and wall friction is neglected (δ = 0). Seismic design uses kh = 0.16, kv = 0. The wall has a 10 ft base and a 2 ft thick stem; passive resistance at the toe is ignored.

Most nearly, the total (static + seismic) active thrust PAE is:

A. 7,403 lb/ft
B. 5,496 lb/ft
C. 14,807 lb/ft
D. 6,219 lb/ft

Given equations / data

> Mononobe–Okabe with δ = 0, β = 0, i = 0: $\theta = \tan^{-1}[kh/(1-kv)]$; $KAE = \cos^2(\varphi-\theta)/\{\cos^2\theta\,[1+\sqrt{(\sin\varphi\,\sin(\varphi-\theta)/\cos\theta)}]^2\}$; $PAE = \tfrac{1}{2}\,KAE\,\gamma\,H^2\,(1-kv)$

Worked solution

1. $\theta = \tan^{-1}(0.16) = 9.09°$
2. $KAE = \cos^2(34-9.09)/\{\cos^2(9.09)[1+\sqrt{(\sin34\cdot\sin24.91/\cos9.09)}]^2\} = 0.381$
3. (static Ka = 0.283)
4. $PAE = \tfrac{1}{2}(0.381)(120)(18)^2 = 7{,}403$ lb/ft

Final answer

7,403 lb/ft

Why the other values are wrong

- **5,496 lb/ft** — used static Rankine Ka
- **14,807 lb/ft** — forgot the ½
- **6,219 lb/ft** — multiplied by (1 − kh) instead of (1 − kv)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.1.5 Pseudostatic Analysis and Earthquake Loads (Mononobe–Okabe), p. 79; GEC 3 §11.3.1

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=w48hBSXrUhY (no creator values copied).

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